

CO1-Assessment Tool-03

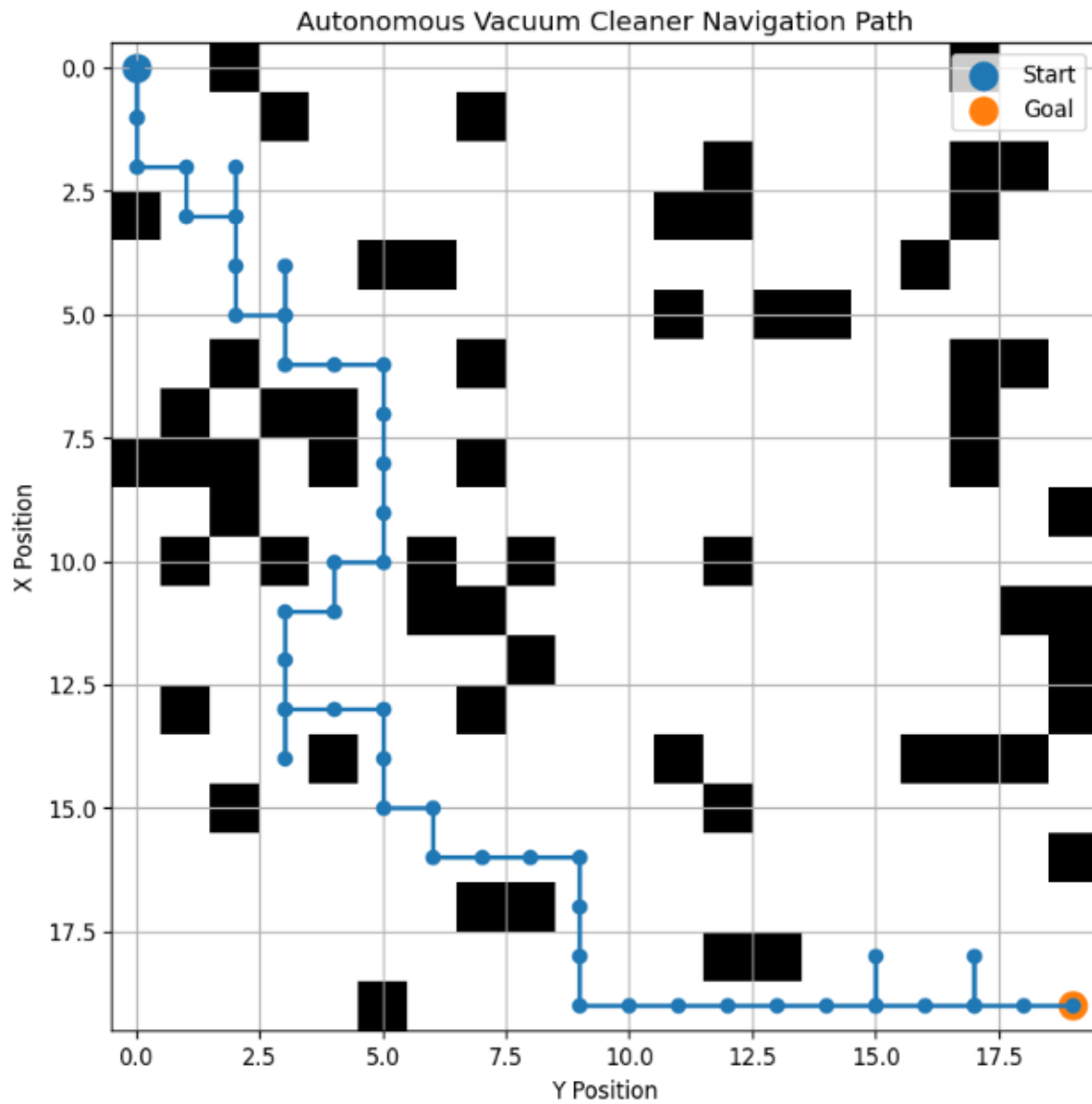
Reinforcement Learning

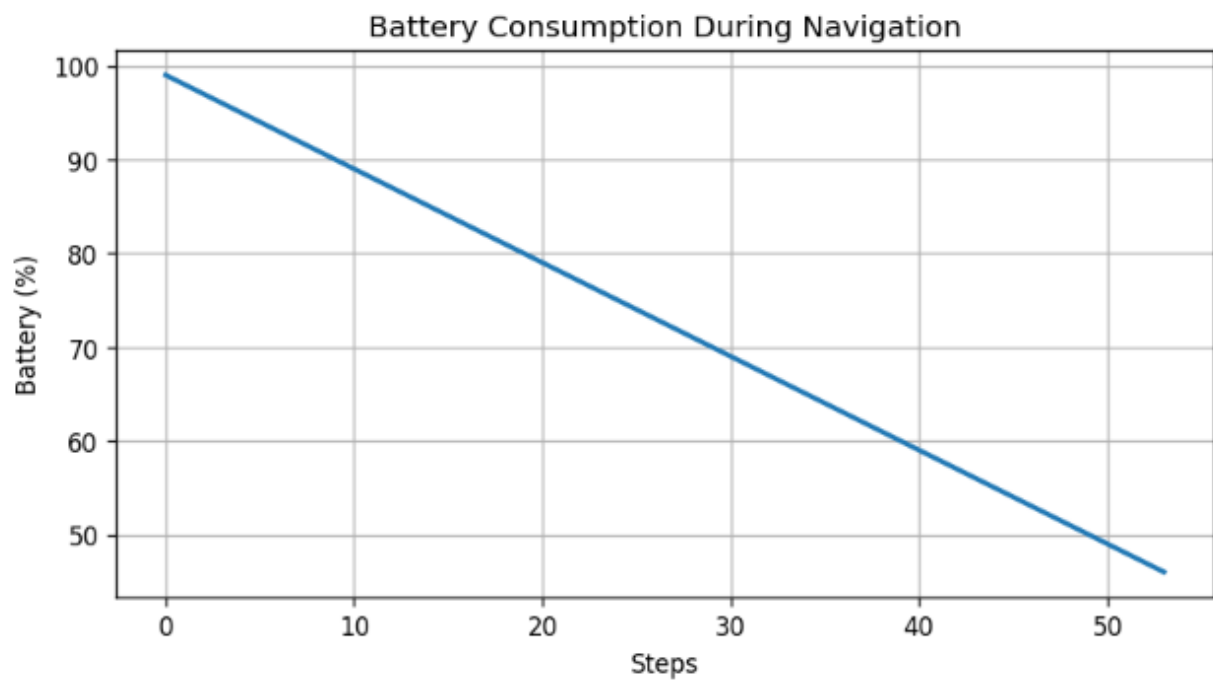
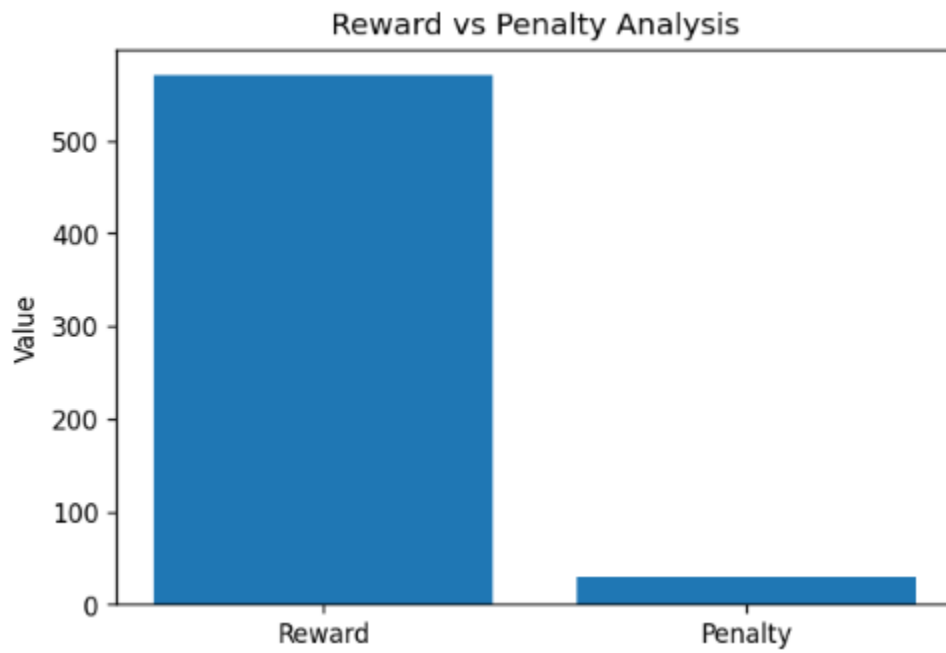
Name: P. Akhila

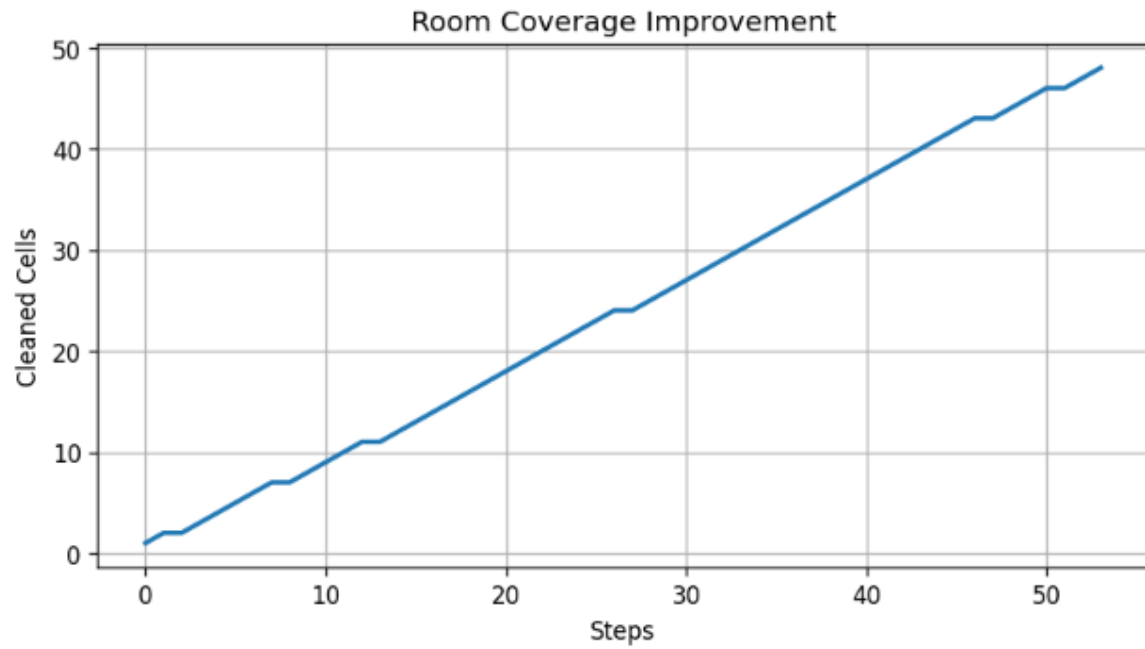
Regno:192472320

Course Code: -MLA0305

Slot-B

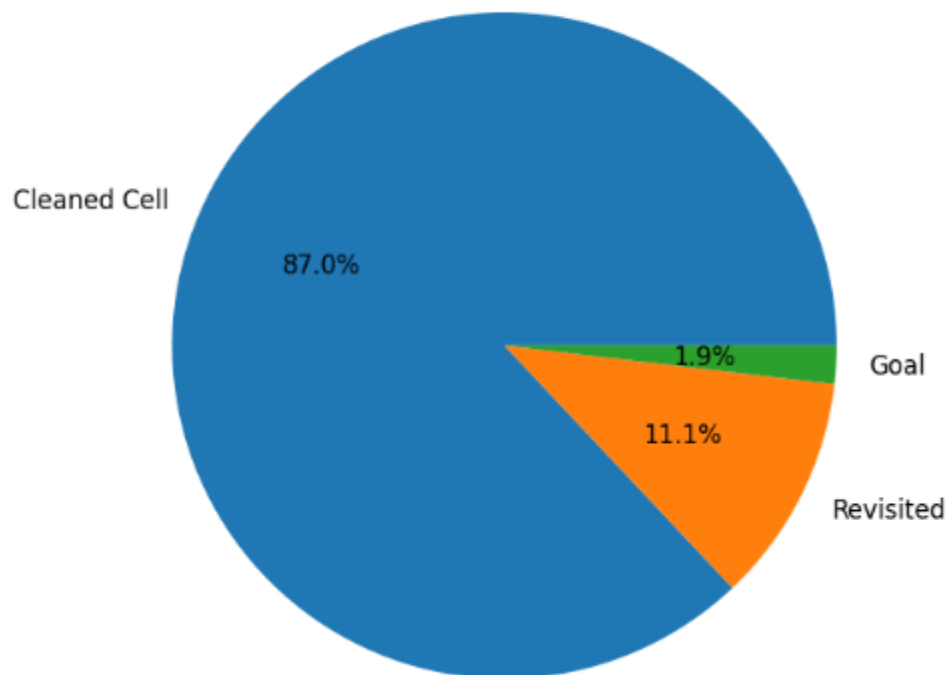


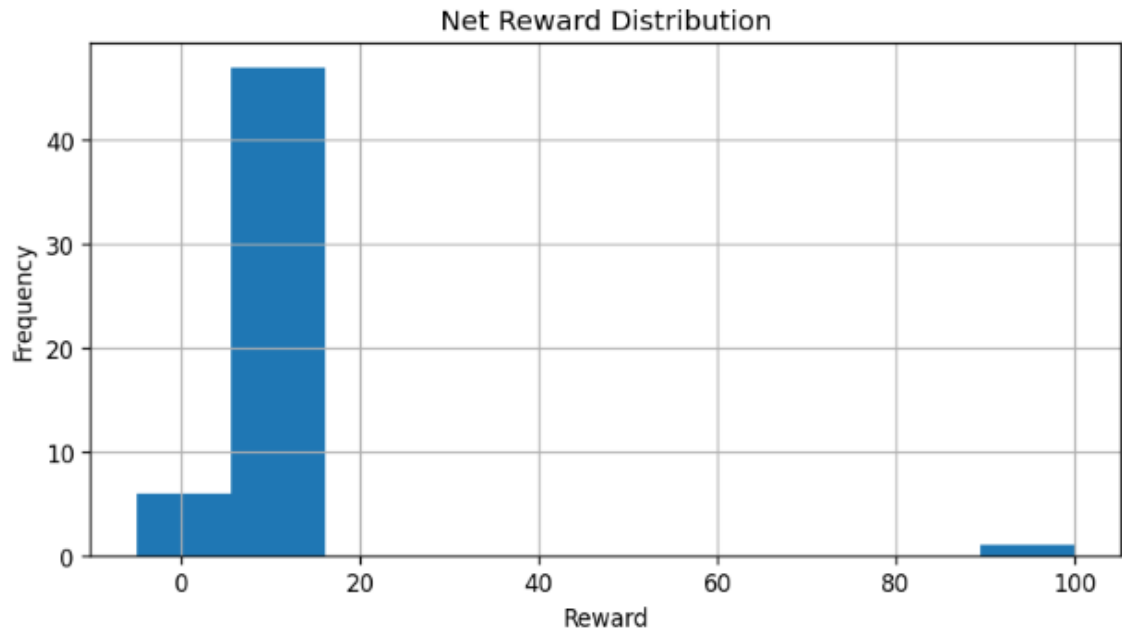




Navigation State Distribution

Navigation State Distribution





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AUTONOMOUS VACUUM CLEANER PERFORMANCE SUMMARY
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	Performance Metric	Value	
0	Environment Size	20 x 20	
1	Total Simulation Steps	54	
2	Coverage Cells	48	
3	Coverage Percentage	12.0%	
4	Total Reward	570	
5	Total Penalty	-30	
6	Net Reward	540	
7	Obstacle Collisions	0	
8	Boundary Violations	0	
9	Revisited Cells	6	
10	Battery Remaining	46%	
11	Goal Status	Reached	

AUTONOMOUS VACUUM CLEANER SIMULATION DATASET

Step	X_Position	Y_Position	Action	State	Reward	Penalty	Net_Reward	Cumulative_Reward	Battery	Coverage
0	0	1	0 DOWN	Cleaned Cell	10	0	10	10	99	1
1	1	0	0 UP	Cleaned Cell	10	0	10	20	98	2
2	2	1	0 DOWN	Revisited	0	-5	-5	15	97	2
3	3	2	0 DOWN	Cleaned Cell	10	0	10	25	96	3
4	4	2	1 RIGHT	Cleaned Cell	10	0	10	35	95	4
5	5	3	1 DOWN	Cleaned Cell	10	0	10	45	94	5
6	6	3	2 RIGHT	Cleaned Cell	10	0	10	55	93	6
7	7	2	2 UP	Cleaned Cell	10	0	10	65	92	7
8	8	3	2 DOWN	Revisited	0	-5	-5	60	91	7
9	9	4	2 DOWN	Cleaned Cell	10	0	10	70	90	8
10	10	5	2 DOWN	Cleaned Cell	10	0	10	80	89	9
11	11	5	3 RIGHT	Cleaned Cell	10	0	10	90	88	10
12	12	4	3 UP	Cleaned Cell	10	0	10	100	87	11
13	13	5	3 DOWN	Revisited	0	-5	-5	95	86	11
14	14	6	3 DOWN	Cleaned Cell	10	0	10	105	85	12

